

THE
Journal of Electricity, Power and Gas

Devoted to the Conversion, Transmission and Distribution of Energy.

VOLUME XX.

SAN FRANCISCO, CAL., MAY 2, 1908

No. 18

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THE Journal of Electricity, Power and Gas

Devoted to the Conversion, Transmission and Distribution of Energy.

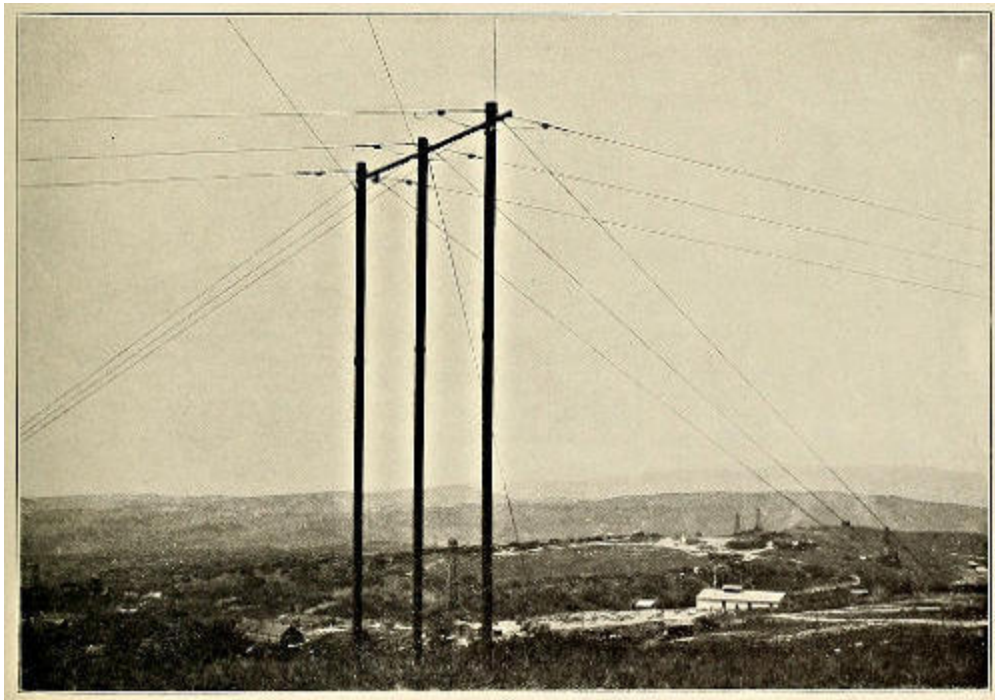
VOLUME XX. SAN FRANCISCO, CAL., MAY 2, 1908 No. 18

LIGHTING SYSTEM OF THE ORCUTT OIL FIELDS.

By CLEM A. COPELAND.

Consulting Engineer, Los Angeles, Cal.

The large and deep-lying oil-sand lakes and subterranean gas works, commencing with the southern rim of the Santa Maria Valley and stretching away for a dozen miles southward toward Santa Barbara, contribute some 14,600,000 barrels of high gravity refining and fuel oil to California's annual production of 40,000,000 barrels.



WEST FRAME OF 2600 FT. SPAN. POWER-HOUSE IN BACKGROUND.

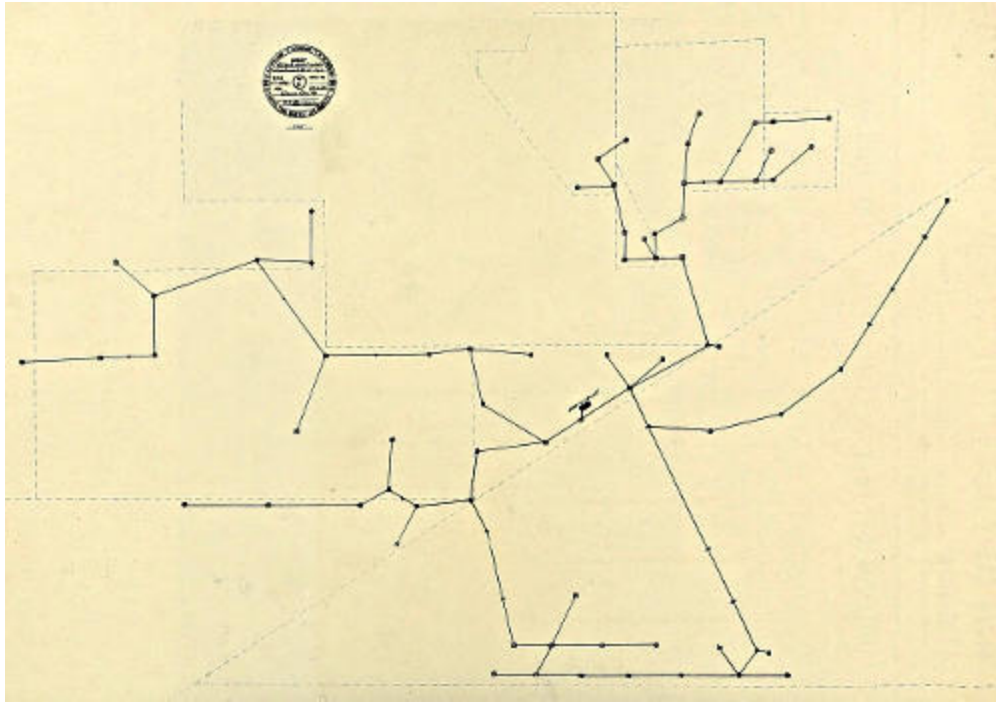
With the assistance of two eight-inch pipe lines 32 miles to Port Harford, a similar line 48 miles across the Isthmus of Panama, and a goodly fleet of vessels, the Union Oil Company scatters this oil from Seattle to San Diego,

and from New York to Japan. Chile also has a share for the working of its nitre beds and its railways.

The little towns of Santa Maria and Orcutt receive with open pipes a tithe of the gas which nature has here stored, and which would otherwise escape the many safety valves, while the steam rig engines have often been run with direct gas pressure from the wells.

This land of gas and gushers is difficult of control, and is always ready to pop off at from 100 to 400 pounds pressure through the many 3,000-foot tubes which puncture its depths. When a new gusher is brought in, it sprays the adjacent hills with a glistening shadow of petroleum and is no respecter of persons or property. One new and frisky fury flowed 12,000 barrels per day, and delivered 4,000,000 cubic feet of gas every 24 hours for four months, gradually dropping to a production of 7,000 barrels, which it maintained for nearly a year, finally diminishing to 3,500, and now, after three and a third years, is still producing 250 barrels per day, having delivered during this time 3,000,000 barrels of petroleum, and enough gas to last San Francisco for three years. This is, with perhaps one exception, the most remarkable well in the history of oil industry, and is widely known as "Hartwell No. 1."

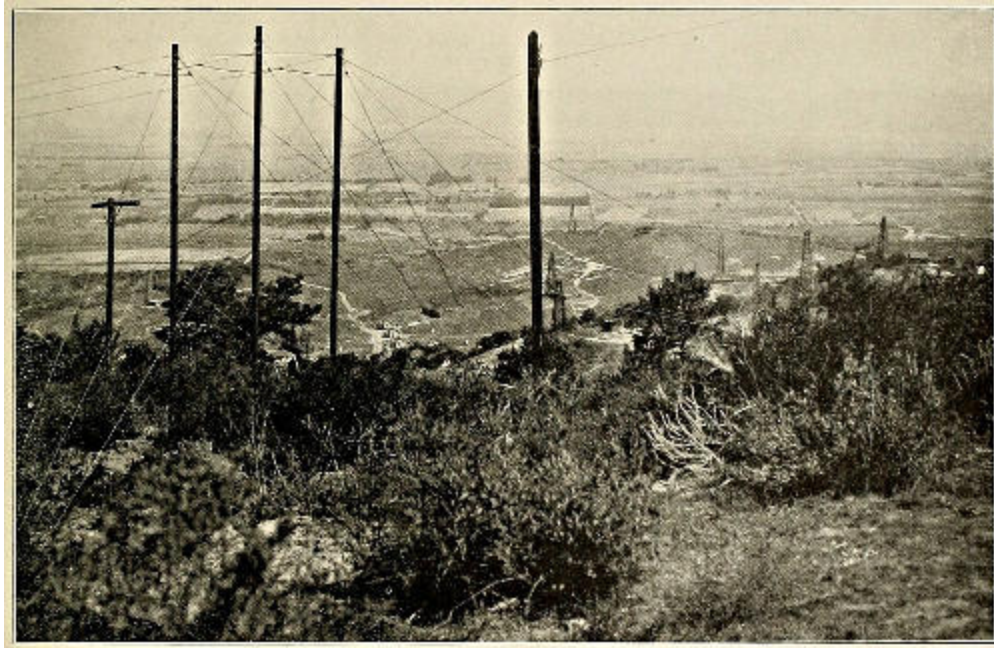
The district contains two groups of wells, one contiguous to Orcutt, and the other near Lompoc. Danger from fire due to the excessive gas pressure of the Orcutt fields is exceedingly great, as evidenced by the burning of four "rigs" in the first two years of its history, during which time there were fifteen wells brought into production. The cost of these "rigs" exceeded the cost of the lighting plant, which is described in these notes, and no fires have since occurred in the sixty wells now producing. The advisability of the plant is therefore quite patent.



MAP OF ORCUTT OIL FIELDS ELECTRICAL DISTRIBUTION SYSTEM.

The lighting system employed in the Orcutt oil fields is perhaps only interesting in illustrating how the methods employed in large undertakings may be used to great advantage in the smaller enterprises to effect a large saving and to simplify conventional methods. The smaller undertakings often afford opportunities of saving a larger percentage in cost and operation in connection with the larger ones. In the present instance, an economy of \$9,000 was made in a system which would have cost \$30,000 if constructed along conventional lines.

As may be seen from some of the views in these notes, the country covered by the system is very hilly and stony, hard sandstone being everywhere prominent. Trees would have interfered considerably over perhaps a third of the line with ordinary construction, and being oak, it would have cost heavily to eliminate them. Although the winters in this section are very bleak and windy, no snow has ever fallen. The attached map shows the wells, which are unusually far apart and scattered over some 5,000 acres of land.



FRAME BETWEEN 1500 AND 1600 FT. SPAN.

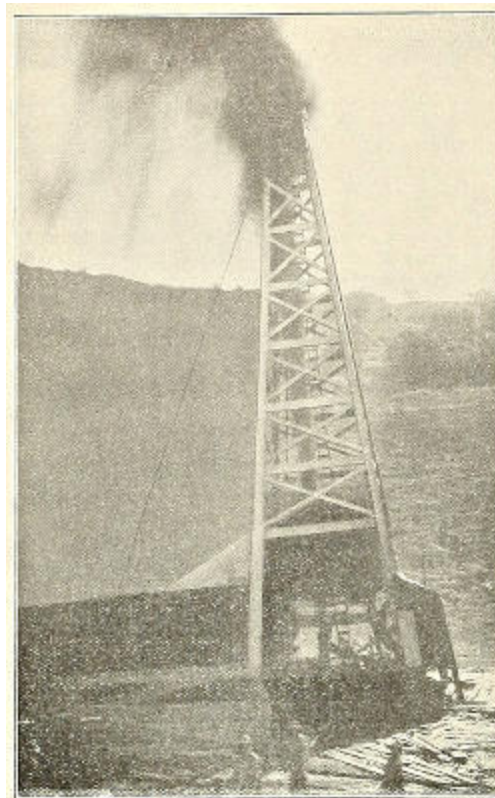
Long span work, employing copper cables, and using the oil-well derricks for support, seemed well to meet these and all other conditions in the most economical fashion. The adoption of long span work effected a large saving in the length of line and wire needed to cover the territory, since air-line routes could be covered in all cases across country, without any care being taken to lay out a pole line which would conform to some general inflexible plan. Moreover, much vertical distance was saved in not having to follow the contour of the country. Incidentally the long span work makes the installation of new pieces of line very easy and simple, and no large amount of material need be kept on hand for expansion purposes.

Where conditions are rapidly changing, as in the present instance, existing lines having to be moved because of the abandonment of wells or re-arrangement as new wells come in amongst the old ones, are easily changed, with but little loss of labor and material. The derricks, 80 feet in height, make ideal supports for long-span construction, and steel frames, heavy enough for the largest size of wire, eventually were made of “scrap” pipe set in cement. Where the derricks were not available, redwood “dead-men” were also used between derricks which were near together, but on opposite sides of rises or hills. The present system of distribution is 72,550 feet, or 13.75 miles, in length, and consists of 70 derricks, 9 frames, and 10 “dead-men,” making the average span about 800 feet. The seven-strand bare

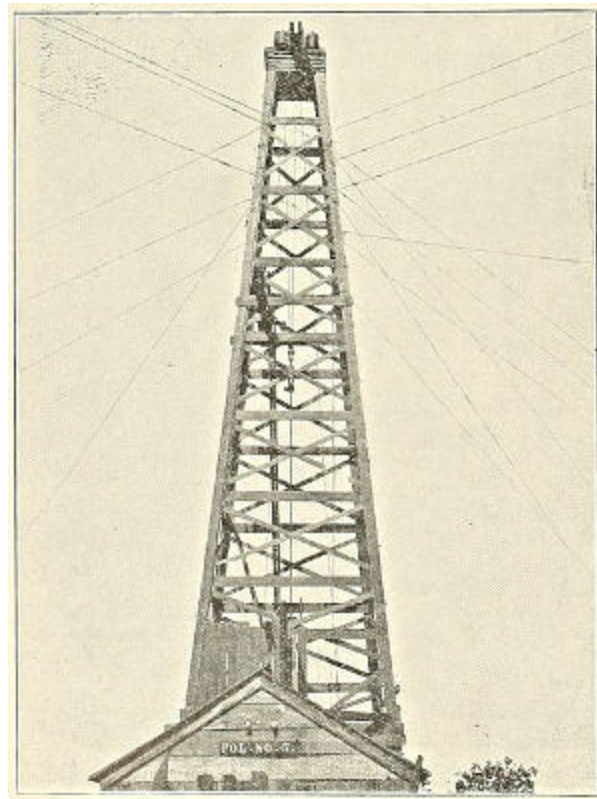
copper cables used in this work were furnished under rigid specifications previously described in the "Journal," by the Standard Underground Cable Company. One will observe from the map and photos that there are many spans 1,500 feet in length, one span of 2,000 feet between derricks and one 2,600 feet from one frame to another. The sags allowed correspond to 60 feet on a 2,000-foot span, and the cables were very small for such work, No. 6 being used for the greatest lengths, which occurs as a neutral on the longest spans. The sizes used are Nos. 2, 4, and 6.

As the cables were suspended high above the ground and good construction was relied upon for safety from breakage, they were used without insulation, and a large saving was thus effected. "Goose-egg" strain insulators, first designed by the writer several years ago, are used to insulate the cables. Copper sleeves were used to splice the cables and to loop them to the insulators.

It is of considerable interest to observe the action of these light cables in a high wind, for even in the most gusty storms there is no whipping action. In the longer spans, the cables hang absolutely parallel and sway in a most deliberate manner from 12 to 25 feet out of line.



HARTWELL NO. ONE.

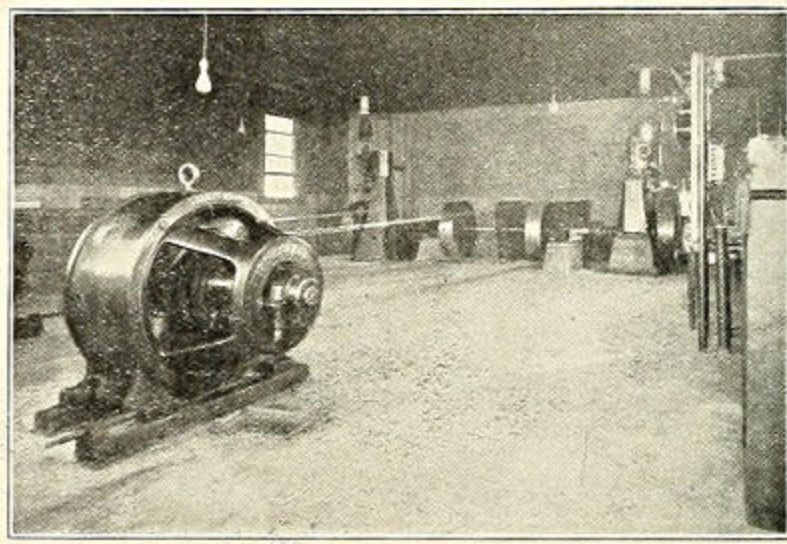


DETAIL OF DERRICK CONSTRUCTION.

Inasmuch as fuel economy is of little importance, since either waste gas or oil can be used, a large drop in the distributing system is permissible, and a radius of three or four miles from the power-house can be economically attained by the use of 210 to 250 volt lamps on the three-wire direct-current system. At present the maximum distance is 255 miles, and four voltages of lamps are employed. The derricks are wired on the two-wire system with No. 14 T. B. W. P. medium, hard-drawn wire, care being taken to keep all wires on the outside of derrick house wherever possible. Although the wires and insulators have been in some cases completely sprayed and saturated with oil from the gushers, no troubles of insulation have been experienced.

At the time this work was started, there was a small plant on the Pinal Oil Company's property near by, where keyed sockets were used until one of the drillers was injured by a gas explosion caused by turning off one of the lights. This, of course, suggested the care necessary to guard against such accidents. In the present installation, double-pole fuses were constructed for each derrick and tank house, by using two weatherproof sockets and Edison plug fuses, all inclosed in a gauze cylinder like a Davy mine lamp. Switches

for tank house and derricks were also inclosed in gauze. In wiring the derricks, sleeves were used for splicing, so that in the whole system no solder nor torch was used. Specially designed heavy wire lamp guards and portables, with wires inclosed in cotton-covered garden hose, are other features of the derrick wiring. When drilling is commenced, the derrick is wired for eleven lights. Tank houses, some fifteen or twenty in number, are wired with a light over each tank.



POWER-HOUSE.

The power-house is of only passing interest, being designed for reliability and minimum first cost, and with the idea of transplanting it to some new location should future conditions dictate. The view shows two 10×10 Shepherd engines, clutched to either end of a shaft, from which two 45-kilowatt, 250-volt, direct-current Westinghouse generators are belted. In case of accident to one generator, a switch on the switchboard converts the 3-wire to a 2-wire system, using the two outside wires as one, and the neutral as the return conductor. Three 48×14 fire-tube boilers supply steam at 125 pounds pressure. The power-house is in a perennially cool location on a hill crest, so that it could be made small and cozy.

The system has been in uninterrupted and satisfactory operation for two years. The only trouble during the time was caused by one wire of one span breaking, due to an imperfection in splicing. The section has long-continued and severe winds, much rain and cold weather, but two winters have developed no imperfections. Although the lights burn all night, no

interruption has been experienced. Reliability is important, since, if the lights failed, there would be a temptation to light candles or lanterns at a critical time.

The system was designed and supervised by Mr. Copeland, of Messrs. Clem. Copeland and F. R. Schanck, consulting engineers for the Union Oil Company. Mr. C. W. Crawley, who is at present electrical superintendent, was foreman of construction.

ALCOHOL vs. GASOLINE FOR POWER.

The Technologic Branch of the United States Geological Survey, under the direction of Mr. J. A. Holmes, has recently completed an elaborate series of tests on the relative value of gasoline and alcohol as producers of power. The tests, over two thousand in number, probably represent the most complete and exact investigation of the kind that has been made, either in this country or abroad, and includes much original research work.

Correspondingly well-designed alcohol and gasoline engines when running under the most advantageous conditions for each, will consume equal volumes of the fuel for which they are designed. This statement is based on the results of many tests made under the most favorable practical conditions that could be obtained for the size and type of engines and fuel used. An average of the minimum fuel consumption values thus obtained, gives a like figure of eight-tenths (.8) of a pint per hour per brake horsepower for gasoline and alcohol.

Considering that the heat value of a gallon of the denatured alcohol is only a little over six-tenths (.6) that of a gallon of the gasoline, this result of equal fuel consumption by volume for gasoline and alcohol engines probably represents the best comparative value that can be obtained for alcohol at the present time, as is also indicated by continental practice. Though the possibility of obtaining this condition in practice here has been thoroughly demonstrated at the Government Fuel-testing Plant, it yet remains with the engine manufacturers to make the "equal fuel consumption by volume" a commercial basis of comparison.

The gasoline engines that were used in these tests are representative of the standard American stationary-engine types, rating at 10 to 15 horsepower, at speeds of from 250 to 300 revolutions per minute, while the alcohol engines were of similar construction and identical in size with the gasoline engines.

The air was not preheated for the above tests on alcohol and gasoline, and the engines were equipped with the ordinary types of constant level suction lift and constant level pressure spray carburetters. Many special

tests with air preheated to various temperatures up to 250° Fahrenheit, and tests with special carburetters were made, but no beneficial effects traceable to better carburation were found when the engines were handled under the special test conditions, including constant speed and best load.

The commercial completely denatured alcohol referred to is 100 parts ethyl alcohol plus 10 parts methyl alcohol plus one-half of one part benzol, and corresponds very closely to 94 per cent by volume or 91 per cent by weight ethyl alcohol (grain alcohol).

No detrimental effects on the cylinder walls and valves of the engines were found from the use of the above denatured alcohol.

The lowest consumption values were obtained with the highest compression that it was found practical to use; which compression for the denatured alcohol ranged from 150 to 180 pounds per square inch above atmosphere.

Eighty per cent alcohol (alcohol and water) for use in engines of the present types would have to sell for at least 15 per cent less per gallon than the denatured alcohol, in order to compete with it. The minimum consumption values in gallons per hour per brake horsepower for 80 per cent alcohol is approximately 17.5 per cent greater than for the denatured alcohol used, or for gasoline. A series of tests made with alcohol of various percentages by volume ranging from 94 per cent to 50 per cent, showed that the minimum consumption values in gallons per hour per brake horsepower increased a little more rapidly than the alcohol decreased in percentage of pure alcohol. That is, the thermal efficiency decreased with the decrease in percentage of pure alcohol. This decrease in thermal efficiency or increase in consumption referred to pure alcohol is, however, comparatively slight from 100 per cent alcohol down to about 80 per cent alcohol. Within these limits it may be neglected in making the calculations necessary to compare the minimum consumption values for tests with different percentages of alcohol.

The nearer the alcohol is to pure, the greater the maximum horsepower of the engine. The per cent reduction in maximum horsepower for 80 per cent alcohol as compared with that for denatured alcohol used, was less than one per cent, but the starting and regulating difficulties are appreciably increased.

With suitable compression, mixtures of gasoline and alcohol vapors (double carburettors) gave thermal efficiencies ranging between that for gasoline (maximum 22.2 per cent) and that for alcohol (maximum 34.6 per cent) but in no case were they higher than that for alcohol. The above thermal efficiencies are calculated from the brake horsepower and the low calorific value of the fuel, which for the gasoline was 19,100 British thermal units per pound, and for the denatured alcohol was 10,500 British thermal units per pound.

As has been previously published, alcohol can be used with more or less satisfaction in stationary and marine gasoline engines and these gasoline engines will use from one and one-half to twice as much alcohol as gasoline when operating under the same conditions. The possibilities, however, of altering the ordinary gasoline engines as required to obtain the best economies with alcohol are very limited; for the amount that the compression can be raised without entirely redesigning the cylinder head and valve arrangement is ordinarily not sufficient, nor are the gasoline engines usually built heavy enough to stand the maximum explosive pressures, which often reach six and seven hundred pounds per square inch. With the increase in weight for the same-sized engine designed to use alcohol instead of gasoline, comes an increase in maximum horsepower of a little over thirty-five per cent (35%), so that its weight per horsepower need not be greater than that of the gasoline engine, and probably will be less.

The work was taken up to investigate the characteristic action of fuels used in internal combustion engines with a detailed study of the action of each fuel (gasoline and alcohol) as governed by the many variable conditions of engine manipulation, design and equipment. These variables were isolated, so far as possible; their separate and combined effects were determined; worked out under practical operating conditions; and led up to the conditions required for minimum fuel consumption. The results show the saving that can be obtained over conditions for maximum consumption, and also establish a definite basis of comparison under conditions most favorable to each fuel. This latter is a point of much commercial interest, and a study of the comparative action of gasoline and alcohol may be of great service in solving some of the general internal-combustion-engine problems where other than liquid fuels are used.

A large number of fundamental tests were necessary in order to clearly define conditions and interpret results. In a way they follow the work conducted by the Department of Agriculture, supplementing to a certain extent, but not duplicating bulletin 191, which gives much data of general value.

Many of the tests of internal-combustion engines have been made, but most of them, especially in this country, were by private concerns, for a specified purpose, and the results are not generally available. Furthermore, as is generally recognized by those familiar with gas, and especially gasoline-engine operation, the conditions influencing engine performance are so numerous and varied as to make the value of offhand comparison very limited and oftentimes misleading, exact comparisons only being possible under identical conditions or with reference to the actual known differences in all conditions that influence the results.

ELECTRICAL CODE REVISIONS.

At the recent meeting of the Underwriters' National Electric Association it was decided that Cooper Hewitt lamps must have a cut-out for each lamp or series, except when contained in a single frame and lighted by a single operation, in which case not more than five lamps shall be dependent on a single cut-out. The regulators must be enclosed in non-combustible cases, and where subject to flyings of lint or combustible material, all openings through the casings must be protected by a fine wire gauze. Moore electric light tubes must be installed so as to be free from liability to mechanical injury or of contact with inflammable material. The high-potential coils and regulating apparatus must be installed in an approved steel cabinet, which shall be ventilated in such a manner as to prevent the escape of flame or sparks in case of burn-out. The apparatus in this box must be mounted on slate, and the enclosing case positively grounded. The supply conductors must comply with the rules governing low-potential systems where such wires do not carry current having a potential of over 300 volts.

Rule 8, section *d*, was amended to apply to auto-starters only, and a new section was added to rule 60 governing the details of rheostat construction. New rules regarding low-potential transformers follow: Oil transformers must not be placed inside of any building except central stations and sub-stations, unless by special permission of the inspection department. Air-cooled transformers must not be placed inside of any building excepting central stations and sub-stations, unless the highest voltage of either primary or secondary does not exceed 550 volts, and must be so mounted that the case shall be one foot from combustible material or separated therefrom by non-combustible, non-absorptive, insulating material, such as slate or marble. Where transformers are placed at a lesser distance, a slab of slate or marble somewhat larger than the transformer must be used, and where the transformer is mounted on a side wall, the slate or marble must be secured independent of the transformer supports, the transformer being supported by bolts countersunk at least one-eighth inch below the surface of the back of the slab and filled.

For wiring electric cranes the following rules were adopted: All wires except bare collector wires, those between resistances and contact plates of rheostats and those subjected to severe external heat, must be approved, rubber-covered and not smaller in size than No. 12 B. & S. Wires between resistances and contact plates of rheostats must conform to No. 4-c, unless the wires are exposed to moisture, in which case the insulation must also be rubber. Wires subjected to severe external heat must have approved slow-burning insulation. All wires, excepting collector wires and those run in metal conduit or armored cable, must be supported by knobs or cleats which separate them at least one inch from the surface wired over, but in dry places where space is limited and the distance between wires as required by Rule 24-h cannot be obtained, each wire must be separately encased in approved flexible tubing securely fastened in place. Collector wires must be supported by approved insulators so mounted that even with the extreme movement permitted the wires will be separated at all times at least one and one-half inches from the surface wired over. Collector wires must be held at the ends by approved strain insulators.

Where the wires are arranged in a horizontal plane above the crane, they must be supported at least every twenty feet if practicable, and separated at least six inches, but if longer spans are necessary, the distance between wires must be increased proportionately, the span in no case to exceed forty feet. If not arranged in a horizontal plane, they must be carried along the runways and must be rigidly and securely attached to their insulating supports at least every twenty feet, and if not arranged in a vertical plane, must be separated at least eight inches.

Where bridge collector wires are over eighty feet long, insulating supports on which the wires may loosely lie must be provided at least every fifty feet. Bridge collector wires must be kept at least two and one-half inches apart, but a greater spacing should be used whenever it may be obtained. Collector wires must not be smaller in size than specified in the following table for the various spans:

Distance between rigid supports. Feet.	Size wire required. B. & S.
0 to 30	6

31 to 60	4
Over 60	2

Collectors must be so designed that sparking will be reduced to a minimum between them and collector wires. The main collector wires must be protected by a cut-out and the circuit controlled by a switch, cut-out and switch to be so located as to be easy of access from the floor. Cranes operated from cabs must have a cut-out and switch connected into the leads from the main collector wires and so located in the cab as to be readily accessible to the operator. Where there is more than one motor on a single crane, each motor lead must be protected by a cut-out located in the cab if there is one.

Controllers must be installed according to No. 4, except that if the crane is located outdoors the wires between resistances and contact plates of rheostats may be rubber-covered or bare or slow-burning if properly supported. If the crane operates over readily combustible material, the resistances must be placed in a fire-resisting enclosure; or, if located in a cab, the cab must be constructed of non-combustible material and sides provided which enclose the cab from its floor to a height at least six inches above the top of the resistance.

The motor frames, the entire frame of the crane and the tracks must be permanently and effectively grounded.

A number of recommendations were made calling for conductive coatings on cables, outlet boxes and fittings in order to secure better electrical contact at all points throughout systems in which they are used.

The following suggested changes in the rules were adopted: The fine print note under rule 2, section a, was amended to read as follows: "Wires from generator to switchboard may, however, be placed in conduit, provided that proper precautions are taken to protect them against moisture and mechanical injury. If lead-covered cable is used no further protection against moisture will be required, etc." Section c of the same rule was amended by inserting the words, "where not in conduit," after the first word. The last sentence of the fine print note in rule 12, section g, was amended to read: "The outer or weather end of conduit is to be provided

with approved devices having wires separated and bushed through porcelain.”

A number of changes were made in rule 24. Section a, with fine-print note, was stricken from the Code; section o was amended by making the first recommendation “a turn of 90 deg., etc.”; section p was amended so as to restrict the number of different circuits in the same conduit to four two-wire or three three-wire; section x was changed by substituting the word *or* after the word *fastenings* for *of*. Rule 24A was also changed somewhat. Section d was amended so as not to prohibit the installation of armored cable without the lead covering in buildings of fireproof construction in locations free from moisture; the word “underground” was omitted from the first line of the fine-print note under section a. A new section was added as follows: “All bends must be so made that the armor of the cable will not be injured. The radius of the curve of the inner edge of any bend not to be less than one and one-half inches.”

Rule 28, section e, was amended so as to exclude flexible cord from show cases as well as show windows; the fine-print note under section g was omitted. A number of minor changes were made in the rules governing the wiring of theaters, principally in inserting the words, “or armored cable,” so as to permit the use of the latter as an alternative to rigid conduit. The other changes have to do with the details of construction, fitting, etc. Among the miscellaneous suggestions adopted were the following: All self-fastening knobs, cleats and supports must be secured by suitable screws; wires in molding must be in continuous lengths from outlet to outlet or from fitting to fitting; sockets or rosettes cannot be used to dead-end a circuit; soap-stone can be used as an alternative for slate or marble; and ends of flexible wire need not be soldered before insertion under binding posts, as called for in rule 14, section c.

Approved Electrical Devices

This department from time to time will contain an illustrated description of all fittings approved by the Underwriters' National Electric Association.

CONDUIT BOX, FLOOR OUTLET.

Iron box, brass floor plate and nozzle. Cat. No. 100. Approved March 20, 1908. Manufactured by

Arthur Frantzen Co., 92 W. Van Buren St., Chicago, Ill.

GROUND CLAMPS.

"Neco" and "Griptite" clamps for rigid conduit, in sizes for ½-inch to 3-inch pipe. "Flexclamp" for Greenfield flexible steel conduits or armored cable, sizes A to E, inclusive. Approved March 20, 1908. Manufactured by

Novelty Electric Co., 50-54 North Fourth St., Philadelphia, Pa.

PANELBOARDS.

Cuthbert Panelboards, 125, 125-250, and 250 V., two and three wire, with double pole knife or snap switches and link, Edison plug or cartridge inclosed fuses. Approved March 20, 1908. Manufactured by

Cuthbert Electrical Manufacturing Co., 105-109 S. Clinton St., Chicago, Ill.

RECEPTACLES FOR ATTACHMENT PLUGS.

Surface receptacles with pull-off attachment plugs. Two-wire, Cat. No. 45,395, 25 A, 125 V; three-wire, Cat. No. 45,490, 25 A, 125-250 V. (For use with approved sub-base only.) Approved March 20, 1908. Manufactured by

General Electric Co., Schenectady, N. Y.

SWITCHES, KNIFE.

Cuthbert Panelboard Switches, 15 A, 125 V, and 25 A, 250 V. Approved March 20, 1908. Manufactured by

Cuthbert Electrical Manufacturing Co., 105-109 S. Clinton St., Chicago, Ill.

CONDUIT BOXES.

“Unilets” cast-iron outlet boxes with threaded openings for ½ to 3-inch rigid conduit. With covers of stamped steel or porcelain, or with porcelain bushings. Types 1-12 and 14. Approved April 13, 1908, for exposed work only. Manufactured by

Appleton Electric Co., 224 Washington St., Chicago, Ill.

FIXTURES.

“Oamco” show window reflector. Cat. Nos. 655, 655A, 655B, 655C. A metal trough lined with glass reflectors and fitted with approved lamp sockets carried on cast-iron arms bolted to iron pipe containing wiring. Approved April 13, 1908. Manufactured by

Overbaugh & Ayres Mfg. Co., 232 South Clinton St., Chicago, Ill.

LAMP ADJUSTERS.

“Gem” lamp Adjusters, styles A and B. A pulley-wheel mounted in iron bracket, wheel controlled by locking mechanism so as to permit the adjustment of a pendant lamp hung from a porcelain knob secured to device. Suitable for use with flexible pendant cord. Approved April 13, 1908. Manufactured by

Gem Mfg. Co., 467 Eleventh Ave., Milwaukee, Wis.

MISCELLANEOUS.

Mercury Arc Rectifiers for converting alternating to direct currents; outfits supplied for A. C. circuits of 110-220 and 330 volts with D. C. current capacities up to and including 50 Amps. (For Telephone Battery Service this apparatus may include G. E. Type A Transformer of unit ratio, to be inserted as insulation between A. C. circuit grounded neutral and ground return of the telephone system.) Compensating reactance case must be mounted on base of slate or over non-combustible insulating material when installed on floor or wall of combustible material. Approved April 13, 1908. Manufactured by

General Electric Co., Schenectady, N. Y.

M. S. Cord Grip. A fibre disc for use in socket and attachment plug caps, rosettes and similar devices, replacing knot in flexible pendant cord. Approved April 13, 1908. Manufactured by

Marshall Electric Mfg. Co., 301 Congress St., Boston, Mass.

RECEPTACLES FOR ATTACHMENT PLUG.

Surface receptacles, with pull-off attachment plug. Three-wire, Cat. No. 45,490, 25 A., 125-250 V., for use only with approved sub-base. Two-wire, Cat. No. 45,395, 25 A., 125 V. Approved April 6, 1908. Manufactured by

General Electric Co., Schenectady, N. Y.

Surface Receptacles, 20 A., 25 V. Cleat concealed and moulding types, Cat. Nos. 5,567 to 5,569, inclusive. Flush Receptacle, 20 A., 250 V., Cat. No. 5,551. Approved April 13, 1908. Manufactured by

Harvey Hubbell, Inc., 35 Organ St., Bridgeport, Conn.

Lang stage receptacle and plug, 125 and 250 V. Base of single piece of hard porcelain or alberene stone. Hardwood plug for stage cable. For use in suitable iron or steel box. Approved April 13, 1908. Manufactured by

J. Lang Electric Co., 116 N. Lincoln St., Chicago, Ill.

Russell stage pocket and plug, Cat. No. 13, 50 A., 125 V. Receptacle with porcelain base mounted in suitable cast-iron box. Plug of red fibre provided with fibre clamp replacing knot in cord. Approved April 13, 1908. Manufactured by

Russell & Stoll Co., 48 Cliff St., New York, N. Y.

RECEPTACLES, STANDARD.

Bryant Receptacles, 3 A., 250 V., Sign, Cat. Nos. 1,700 and 46,749. Cleat Cat. Nos. 9,402, 9,403, 921, 1,011, 1,123, 50,715, 11,221, 28,795, 58,949, 58,300, 58,301. Concealed, Cat. Nos. 50,744, also 9,447, fusible 2 A., 125 V. Moulding, Cat. Nos. 42,453, 58,302 and 58,950. Conduit box. Cat. Nos. 9,514 and 9,397. Rosette receptacles, cleat and concealed types, fusible, 2 A., 125 V., Cat. Nos. 9,434, 9,436, 9,438, 9,404, 9,405 and 9,406. Approved April 2, 1908. Manufactured by

Bryant Electric Co., Bridgeport, Conn.

Porcelain shell, keyless, 3 A., 250 V. Cleat type, Cat. Nos. 28,794, 28,795 and 11,221. Concealed type, Nos. 50,744 and 50,717. Conduit boxes, Nos. 9,397, 40,537, 49,354 and 9,514. Sign receptacle, No. 46,627. Wall sockets, brass shell, key, 50 C. P., keyless. 3 A., 250 V. Concealed base, Nos. 9,184, 27,743, 29,404, 9,185, 27,743 and 29,405. Angle base, Nos. 50,753, 28,721, 29,406, 50,755. 28,722 and 29,407. Approved April 13, 1908. Manufactured by

General Electric Co., Schenectady, N. Y.

THE JOURNAL OF ELECTRICITY, POWER AND GAS

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VOL. XX MAY 2, 1908 No. 18

EDITORIAL.

MANUFACTURING OPPORTUNITY.

The first half century of Western development was accomplished without the aid of cheap fuel. This lack was really a blessing in disguise, for it applied the needed spur that finally forced the utilization of its protean water powers. But with the development of its latent oil resources there became available a fuel whose cheapness and convenience should enable the West to compete in the manufacturing field.

There has been no factor of greater use and with less recognition, in this attainment, than the aid given by the engineer. The first and most pressing problem was the taming of the furious force of newly tapped gushers, so as to regulate and control their flow with reference to reservoir capacity. But so irresistible were some that they could no more be restrained than can the waves of the ocean. Millions of barrels flowed to wanton waste. Earthen barricades were hastily built around open pools, and considerable oil was thus preserved, only to be flooded into an unready market, whose demand was not sufficient to absorb the sudden oil supply. Nor were the existing means of transportation suitable.

Here, indeed, was work to be done. How quickly a market was created is indicated by the fact that in 1906 the local consumption was greater than the production. This was due to an accumulation caused by low prices, the average being less than twenty-five cents, and the minimum ten cents. A fleet of tank steamers and a pipe line across the Isthmus of Panama soon gave access to the Atlantic, and Pacific possibilities were vigorously developed, so that the demand has already doubled this price.

With regard to the crude oil, boilers and furnaces have been reconstructed so as to efficiently use the new fuel. Radical changes were necessary in order to provide increased air capacity and new means of air distribution. The very best of our engineering talent have been so successfully concerned with this problem that fuel oil is now used in nearly every steam plant on land and sea, for both stationary and locomotive use. Petroleum-enriched water gas is much superior in cost and quality to the illuminant it is

displacing, and now requires considerable oil. Smelting with oil will soon be an accomplished fact in spite of many difficulties, and oil engines such as the Diesel also utilize the crude.

Following its utilization as fuel, there came its varied applications as a refined product. Gasoline and engine oils were separated for use in the various types of explosive engines, lubricating oils were distilled to make the running easier, and from the residue, asphalt was taken to meet the good roads movement. Many other varied uses have been developed for this product, including its application to roofing and weatherproofing. Chemists have found that most of the California oil, as well as that from the Texas fields, contains an asphaltum rather than a paraffine base, which thus distinguishes them from most of the Eastern oils.

Coincident with this increasing use of oil during the past decade have been the improvements in the method of carrying it. The great expense of reaching this engineer-made market demanded that cheap transportation be provided. Long pipe lines have been built, which pour the oil directly into the refinery or into ships that take it to a foreign market. One nearly three hundred miles long was constructed, but proved inoperative on account of the great viscosity of the oil. But even this difficulty has been met and overcome by an entirely new principle that promises to solve the problem, and forms one of the interesting stories the "Journal" has yet to tell.

Not less important than the engineering problems already outlined is that detailed in this issue by our friend, Mr. Clem. A. Copeland. Disastrous fires have consumed millions of gallons, and any means that lessens this needless waste is welcomed. Incidentally it illustrates the importance of studying other jobs in relation to their possible application to new problems. The adoption of long-span work required courage, which has been justified by the results. It is not our province to discuss the ethics of competitive struggles which have characterized the history of oil wherever developed in large quantities in this country. But, as long as competition endures, stagnation is prevented, and it is undoubtedly due to this stress that so much engineering ingenuity has resulted.

COMMERCIAL DAY AT THE NATIONAL ELECTRIC LIGHT ASSOCIATION

Mr. Geo. W. Williams, who is one of the best known commercial men in the country, and Mr. Frank B. Rae, of Selling Electricity, are joint editors on what will probably be one of the very interesting features of that particular part of the convention, namely: "An illustrated talk on the methods of creating demand for electricity." This will include stereoptican views showing in detail the progress of the outline and sign lighting in large and small cities.

PERSONAL.

C. K. King, vice-president of the Ohio Brass Company, is expected in San Francisco this week.

P. H. Coolidge has come from Chicago to take the management of the Western Electric Co. of San Francisco.

W. A. Blair succeeds Mr. R. L. Van Valkenberg as assistant treasurer of the Western Electric Co. Mr. Van Valkenberg will go East on May 15th.

Alonzo Gartley, General Mgr. Hawaiian Electric Co., Honolulu, is in San Francisco as a member of the Hawaiian Governor's staff, on the way East to attend the Roosevelt conference.

W. I. Otis, who has been associated with the Western Electric Co. for the past five years, severs his connection with the company on May to open an office at 111 New Montgomery St., is the representative of several responsible Eastern manufacturers.

Tracy E. Bibbins, San Francisco manager Supply Department General Electric Co., has recovered from the shock and minor injuries received in the recent wreck of the "Owl," and is receiving the congratulations of his friends on having escaped without serious injury.

James D. Schuyler, hydraulic engineer, of Los Angeles, Cal., has been appointed member of a board of engineers to investigate and report on a power plant in Japan for an English syndicate to operate the street railways in Tokio and Yokohama, and to furnish current for lighting and power of those cities.

OBITUARY

Mr. Samuel B. Rawson, president of the Dean Electric Co., died Thursday, April 9, 1908, at Elyria, Ohio.

TRADE CATALOGUES.

Holophane Reflectors for Gem, Meridian, Tantalum and Tungsten lamps are illustrated and described in Bulletin No. 6 from the Holophane Company of New York City.

Bulletin No. 21 from H. Krantz Mfg. Co., 160-166 Seventh St., Brooklyn, N. Y., illustrates and describes Standard and Water-Tight Boxes for all electrical installation purposes. This line includes floor, wall, ceiling, elevator, receptacle, switch, conduit and junction boxes, as well as boxes with plugs and receptacles for either wood, concrete or parquet floors or marine installations.

The General Electric Company, Schenectady, N. Y., in Bulletin No. 4,576, describes the Type F, Form K-3 line of oil switches for panel installation and remote control, on systems of 4,500 volts or less. The object of Bulletin No. 4,578 is to describe the essentials of the various standard controllers that are manufactured for railway service, with special reference to the operating conditions for which each type is suited. Among the controllers described are Type B, which include the necessary contacts and connections for electric braking, Type K for series parallel operation of the motors, Type L, also in the series parallel class, but which completely open the power circuit when changing from series to parallel; Type R, which are designed to control the motor speeds by means of resistance only, and a brief outline of the Sprague-General Electric Type M control system. The General Electric CQ motor is described in detail in a 16-page bulletin. This motor is for direct-current circuits, and is made up to 20-horsepower in size, and for voltages, of 115, 230 and 550. The application of the motor to linotype equipments, ventilating outfits, machine tools, etc., is also outlined. The extensive tables of dimensions, capacities, etc., in the bulletin, will be found very useful in preparing specifications.

CIVIL SERVICE EXAMINATIONS.

Switchboard Attendant (Male).

The United States Civil Service Commission announces an examination on May 6, 1908, to secure eligibles from which to make certification to fill a vacancy in the position of switchboard attendant (male), \$900 per annum, United States Military Academy, West Point, N. Y., and vacancies requiring similar qualifications as they may occur in any branch of the service. The examination will consist of letter-writing, practical questions, and experience. Applicants for this examination should be practical electricians. They should state accurately in their applications what experience they have had in the handling of both alternating- and direct-current switchboards and alternating- and direct-current generators; also experience with various meters used in measuring high-voltage currents, rheostats, transformers, and other apparatus used in a power house.

An examination will be held on May 6, 1908, to fill a vacancy in the position of assistant engineer, \$900 per annum, office of the Attorney-General, Washington, D. C.; a vacancy in the position of engineer, \$1,020 per annum, Freedmen's Hospital, Washington, D. C.; and vacancies requiring similar qualifications as they may occur. Applicants who have had experience in plumbing should so state in their applications, as such experience is required for the position in Freedmen's Hospital. The examination will consist of letter-writing, practical questions in mechanical and electrical engineering (comprising the construction and operation of the heating plant and electric lighting and elevator machinery in first-class public buildings), and experience in mechanical and electrical engineering work.

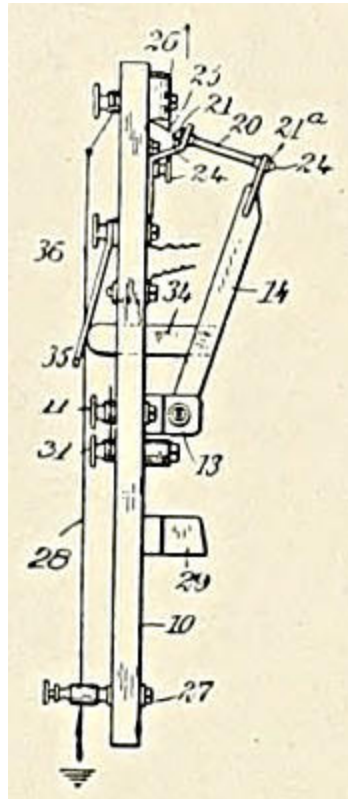
GAS ENGINEERING.

In order to meet the constantly increasing demand for training in gas engineering, four courses have been established at Cornell University. First, a course of lectures on the general theory of gas engines; second, a course of lectures on gas engine design; third, a drafting room course in gas engine design; and fourth, a lecture course which treats of the engineering problems involved in the conversion of various solid and liquid fuels into gas fuels, and in the transmission of gas fuels. This course discusses the different gas making processes and gives descriptions and studies of designs of apparatus used. The object of these four courses is to give to the student taking them the fundamental ideas of modern gas engineering.

The gas engine laboratory, which was moved into one of the small buildings in the court after the building had been rendered fireproof by concrete walls and floor, has proved a very satisfactory addition to the laboratory and will doubtless produce much valuable scientific information. It is now equipped with examples of every important type of gas engine which has been produced since the time of Brayton. A producer gas plant is being installed and will soon be in practical operation. Through the kindness of John Wilkinson, M. E., Cornell 1889, chief engineer of the Franklin Co., a four-cylinder Franklin automobile engine has been presented.

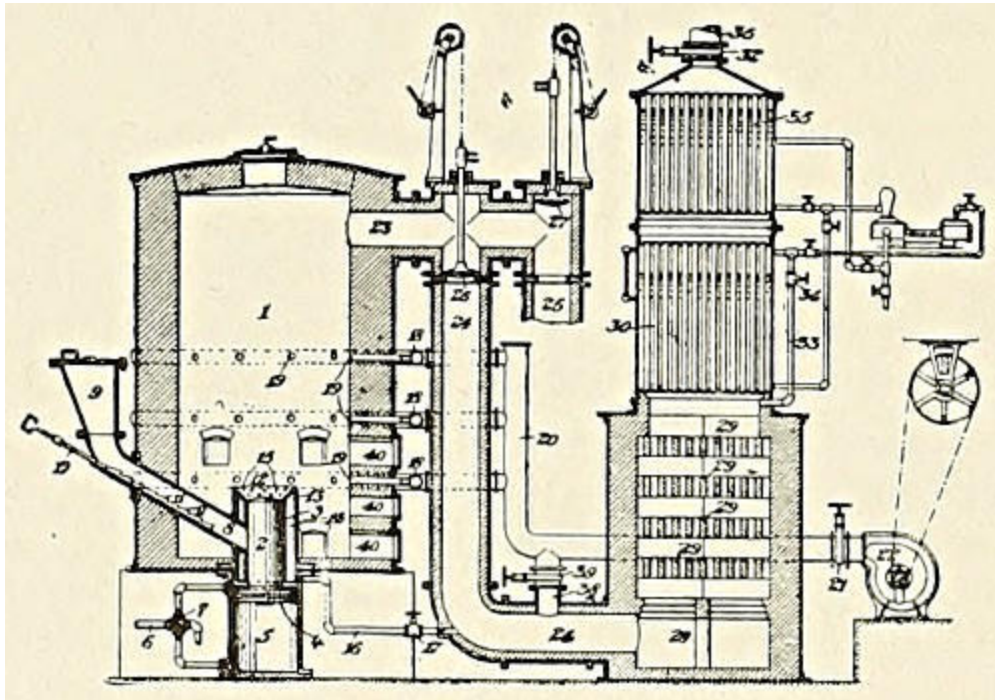
PATENTS

CUT-OUT. 884,978. John H. Booth, Frank E. Blausey, and Arthur M. Smith, San Pedro, Cal.



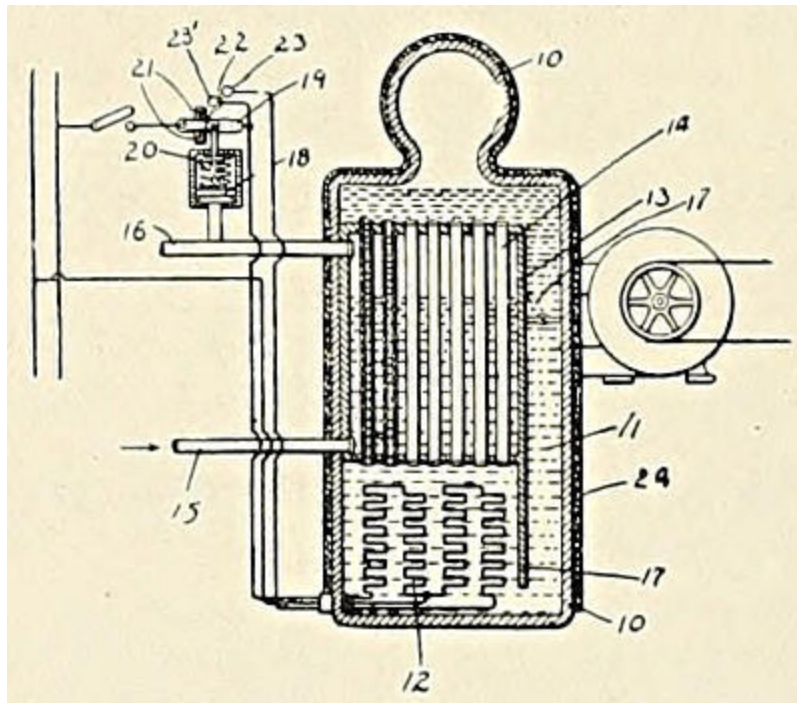
A cut-out comprising a base having a line terminal and an instrument terminal, a drop lever pivotally mounted on the base, line terminal being connected with the pivotally mounted drop lever, lever having its free end provided with a fork, a bracket to which the instrument terminal is connected, bracket being also provided with a fork, a fuse resting in the forks, and supporting the drop lever, and a carbon block having a ground connection connected with the cut-out adjacent to the end of the fuse which rests in the fork of the bracket.

PROCESS OF MAKING GAS. 884,655. Alexander M. Gow, Edgewood Park, Pa., assignor, by mesne assignments, to the Westinghouse Machine Company.



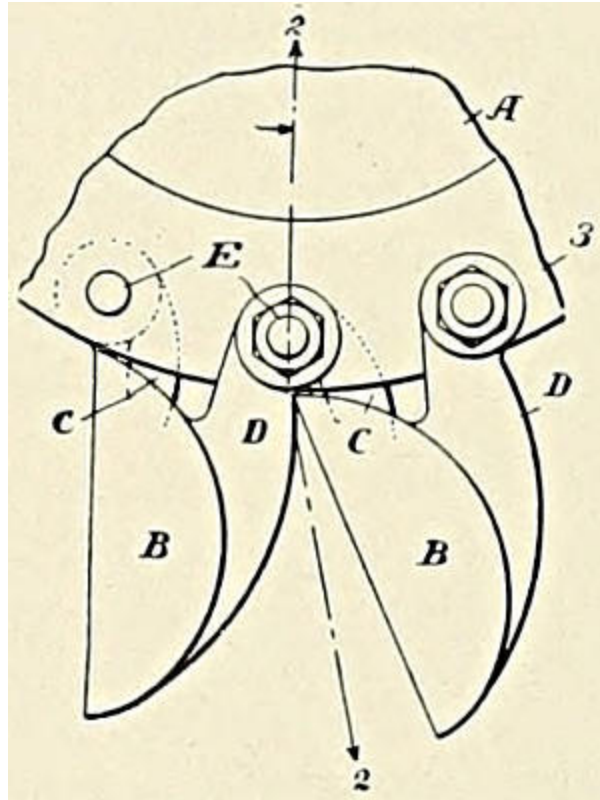
The process of making gas, which consists in blowing to incandescence the exterior portion of a body of fuel, forcing fresh fuel into the interior of the body, utilizing the sensible heat of the incandescent fuel in the destructive distillation of the fresh fuel, introducing steam into the interior portion of the body of fuel and causing it to pass, first through the fresh fuel and then through the heated exterior portion of the fuel bed.

ELECTRIC HEATER. 884,540. Elihu Thomson, Swampscott, Mass., assignor to General Electric Company.



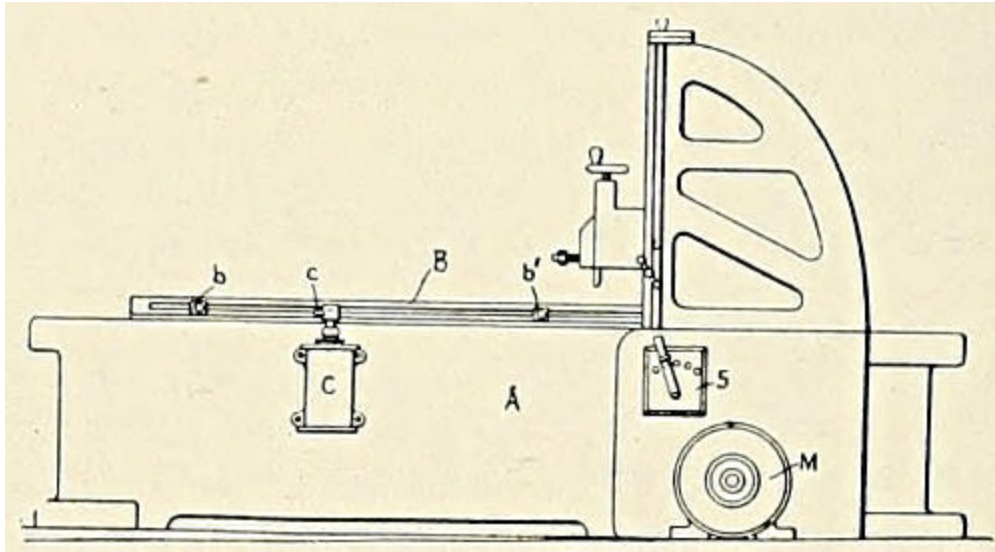
An electric heater comprising a receptacle, containing oil having a high flashing point and means for circulating the same, a resistance conductor immersed in the oil and adapted to heat the same, and a heating system partially within the container and arranged to receive its heat from the circulating oil.

IMPACT WATER-WHEEL. 884,907. William A. Doble and Frederick Gfeller, San Francisco, Cal.



The combination with the body of an impact wheel, of buckets each having at separated points, one in advance of the other, a central and two parallel perforated ears, the perforations of the central ear of each bucket axially in line with those of the parallel ears of the contiguous bucket, and bolts passing transversely through perforations and through the body.

SYSTEM OF CONTROL FOR ELECTRIC MOTORS. 884,541.
Leonard A. Tirrill, Lynn, Mass., assignor to General Electric Company.



The method of operating a compound-wound motor, which consists in supplying current to the armature and field windings, cutting out the armature and connecting the series field winding to the source of current supply, connecting the armature in reverse relation to the series field winding, and then inserting a high resistance in series with the shunt field winding.

THE PASSING OF A PIONEER.

In our advertising columns this week, announcement is made of the retirement from active business of the California Electrical Works of San Francisco and the continuance of that business under the name of the Western Electric Company, which has directed its operation since it assumed control in 1901.

With this change there passes into history a name so closely identified with the pioneer days that to write the history of the California Electrical Works is to write the history of the electrical business of the early days of California.

The year 1871 witnessed the incorporation of the Electrical Construction and Maintenance Company, afterward the California Electrical Works. The officers were George S. Ladd, president, John G. Ayres, business manager, and S. B. Field, electrician and secretary. Their location was on the top floor of a small frame building at the corner of Montgomery and Jackson Streets, San Francisco.

This company absorbed all the business formerly handled by outside electrical enterprise and acquired by purchase the interests of Lundberg & Marwedell, who had been carrying on a limited business in the manufacture of telephone apparatus and supplies, including the manufacture of all telegraph instruments used by the Western Union Telegraph Company on the Pacific Coast.

Previous to the incorporation of the Electrical Construction and Maintenance Company, electrical development on the Pacific Coast had been entirely confined to the operations of the California State Telegraph Company.

In the year 1863, three telegraph circuits entered the city of San Francisco, these being composed of No. 9 iron wire supported on redwood poles and insulated with hook insulators; but two of these circuits could be worked at the same time, the "cross fire" between conductors rendering a separate pole line for each conductor a necessity. In those days every

trouble on the line was attributed to the fogs which were then as now very dense at certain periods of the year.

In the year 1865 the fire alarm telegraph was introduced in San Francisco, and Mr. Field was one of the operators of this system. The following year, 1866, the first private telegraph line was constructed between the offices of Kelly, Hewston & Company and their refinery about a mile away. This circuit was made of baling wire—about No. 14 plain iron—with a joint every 150 feet. The instruments, battery and dial (“A. B. C.”) were from A. T. & J. N. Chester & Company of New York. The success of this line was so immediate that quite a demand sprang up for such installations, numbers of which were constructed and installed by the Electrical Construction and Maintenance Company under the supervision of Mr. Field. It may be of interest to know that these instruments cost \$125 each in greenbacks, which were then rated at about 50 to 80 cents on the dollar, and were sold in San Francisco for \$250 each in gold.

Housetop line construction cost \$125 per mile and pole line construction about \$250 per mile.

About 1874, the American District Telegraph System was introduced in San Francisco, the promoters being the various stockholders of the Electrical Construction and Maintenance Company, although the business was carried on under a separate incorporation.

On June 14, 1876, the company consolidated with Mr. Paul Seiler and Dr. Hirsch and incorporated under the name of the California Electrical Works, changing their location to the third floor of a brick building on Sutter Street.

In the year 1877 the telephone was introduced on the Pacific Coast, the first exchange being opened in the District Telegraph office on Sansome Street, San Francisco. In the summer of 1878 the first long-distance telephone line was constructed for the North Bloomfield Gravel Mining Company and various other adjacent claims commenced at French Corral in Yuba County. It extended to nearly the summit of the Sierra Nevada Mountains, the total length of the line being about sixty miles with twenty-four stations. This line was installed by the California Electrical Works, the principal work being done by Mr. John O’Neil. In this connection it will no doubt be of interest to know that he is still in the employ of the company

and we are indebted to him for much of the information contained in this article.

During these early years the California Electrical Works wired most of the principal buildings and residences in San Francisco, including the old Palace Hotel, the James B. Haggin residence, the Goad residence, the Concordia Club, General Colgan's residence, the Mark Hopkins' Institute, the Leland Stanford residence, the Chas. Crocker residence and many others. These names represent many of the most prominent early pioneers in California. Most of the wiring was for call bells and the wire used on the first installations was insulated with cotton braid and afterwards varnished, the work being done in the shop of the California Electrical Works.

The first drop annunciator on the Pacific Coast was installed by the California Electrical Works in the office of the Grand Hotel which was directly across the street from the old Palace Hotel. This annunciator remained in service until the earthquake and fire of 1906.

In 1877 the company installed the first electric light power station on the Coast. The old California Theater and Auction House, once so famous, was nightly illuminated with "Jarblokoff" candles. During the year 1878 the company obtained from Mr. Thos. A. Edison the first generator for incandescent lights seen in the West, this machine being one of the historical bi-polars with a capacity of about 160 lights. It was installed in the H. S. Crocker building on Bush Street, as an exhibit, and attracted a great deal of attention. The sockets for the lamps were very cumbersome and had no keys and there was only one switch on the line, this being placed on the frame of the generator. Shortly afterwards, this machine was installed as a permanent exhibit in the old Mechanics' Pavilion on Larkin Street, and as a result the California Electrical Works installed quite a number of them during the next two years.

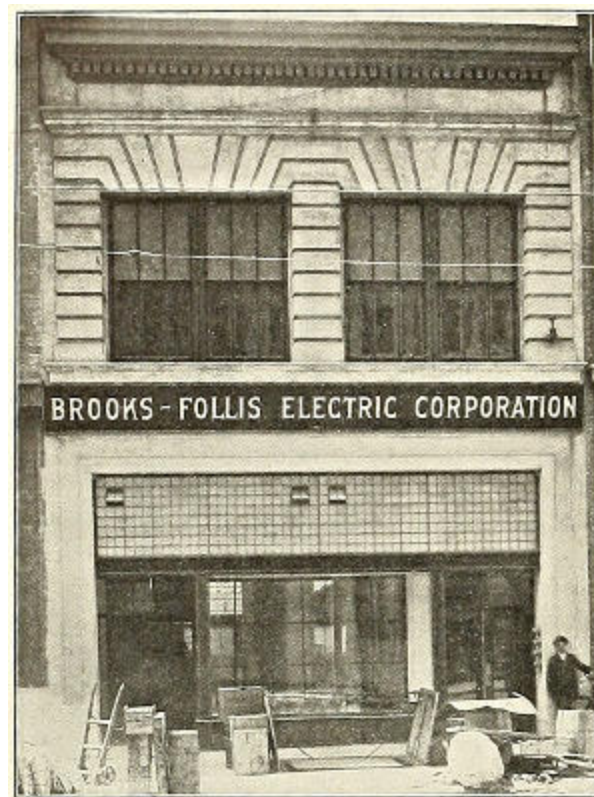
In 1886 the business was moved to 35 Market Street, where better accommodations and increased floor space were obtained. In 1890 they again moved to 409 Market Street, where, owing to the increased volume of the business, they occupied three floors. On December 16, 1901, the Western Electric Company acquired the controlling interest in the company and located at 547 Mission Street, where three floors and a basement were occupied for offices and stock. They remained in these quarters until the

completion of their present building at 642 Folsom Street, where they moved in January, 1906, three months before the catastrophe which visited San Francisco.

Farewell to the California Electrical Works. Welcome to the Western Electric Company.

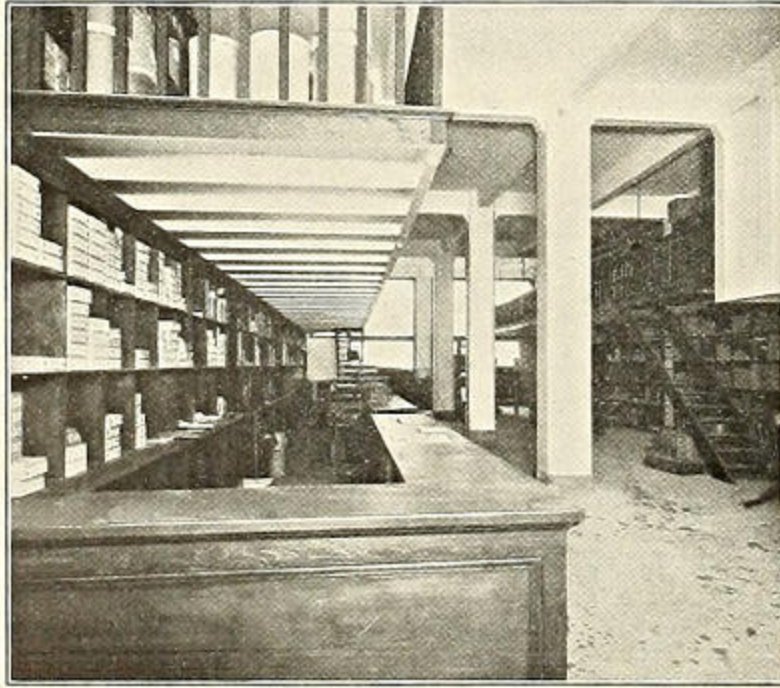
BROOKS-FOLLIS AT HOME.

The accompanying illustrations show the new home of the Brooks-Follis Electric Corporation, 44-46 Second Street, San Francisco, Cal. Mr. Frank Fowden, President of the company, has clearly demonstrated what can be accomplished through perseverance, will and optimism in San Francisco's future.



EXTERIOR.

At the time of the fire in 1906, this firm was located on Mission Street. The business was apparently demolished, but April 24th found the Brooks-Follis Electric Corporation bobbing up serenely at 563 Thirteenth Street, Oakland, where shipments were made as rapidly as material could be obtained.



INTERIOR.

In July, 1906, a corrugated iron building at 212-214 First Street, became the temporary San Francisco home of the firm, where all orders were handled to the best possible advantage under the conditions existing at that time. These quarters were acceptable while San Francisco was being settled and the business growing.

At the present central location all the electrical men from the interior can take a Market Street car and drop in at 44-46 Second Street, take a seat in Mr. Fowden's office, read the "Electrical Journals," use the telephone and make themselves quite at home.

TURBINE ACTIVITIES IN THE FAR EAST.

No less than ten machines, aggregating 25,000 horsepower, are included in a large shipment of Westinghouse turbo-electric power equipment from East Pittsburg to the Far East. Most of these machines will go to Japan for the equipment of railway, lighting, and manufacturing plants.

One of the first machines to be put in service will be a 1,500-kilowatt turbine unit for Manila, to be installed in a station with four other machines of like construction put into service several years ago. Past experience with these machines has resulted in the recent extension. It will be recalled that this railway system was engineered and constructed by the American engineering firm of J. G. White & Co.

Hardly second in importance is the large turbine station of the Osaka Electric Company, Osaka, Japan, now building. This will be one of the largest power stations in Japanese territory, and will contain, for the present, 15,000 kilowatts in five units. Three of these machines are now being shipped from East Pittsburg. The remainder will follow as fast as they can be built and tested. The Osaka installation is under direct charge of Messrs. Takata & Co., of New York and Tokio.

In the strictly manufacturing field, there are two installations in process of erection, for the Imperial Steel Works of the Japanese government and the ship yards of the Hakkaido Tanko Steamship Company. Two 500-kilowatt Westinghouse-Parsons turbo units will comprise an initial installation in each of these plants. This gratifying reception of American motive-power machinery in the Far East, especially Japan, may be regarded as an index of future operations where government inspection is exceedingly rigid and is exercised along lines much more detailed than in this country.

The General Electric Company has obtained very satisfactory results with the tungsten lamp. They have shipped over 75,000 to all parts of the country, and the breakage in shipment is below one and one-half per cent. They state that they are issuing a new bulletin covering tungsten lamps, both series and multiple, and have a large production, good stocks, and are in position to make prompt shipment, particularly of the 100-watt and all types of tungsten lamps which they have standardized.

TRADE NOTES

The American Automatic Telephone Co., formerly of Rochester, N. Y., and the Select Telephone Mfg. Co., formerly of Springfield, Ohio, are now located in their new plant at Urbana, Ohio, under the name of the American Automatic Telephone Co.

NEWS NOTES

POWER AND LIGHT.

Meadow Lake, Wash.—The Washington Water Power Company will build a power-house.

Eugene, Ore.—It is reported that the Willamette Valley company which supplies Eugene with electric light and power is soon to install an entirely new equipment of engines and dynamos in the generating plant at this place.

Republic, Wash.—The county commissioners granted to Arthur Phillips and H. D. Merritt franchises, each for a period of thirty-five years, to lay water mains from Deer Creek to the town of Orient, water pipes throughout the town, and for a pole line and wiring lighting the town and furnishing heat and power.

Twin Bridges, Mont.—Henry Pankey, superintendent of the Easton and Pacific mines, above Virginia City, has returned from California and will at once push matters connected with the erection of an electric power plant at Blaine Springs, which is intended to furnish electric power for the lighting of the county seat.

New Westminster, B. C.—Managing Director Buntzen, of the British Columbia Electric Railway, held a conference with the city council recently to discuss the necessity of erecting a new dam at Lake Coquitlam. Mr. Buntzen agreed that it was necessary, the present one being but a temporary affair. He will recommend the erection of a dam of the best construction to be put up this year.

Cranbrook, B. C.—The mining and general industries of southeast Kootenai will be vigorously stimulated by the fact that Wisconsin capital has now been secured to put through and fully equip the undertaking of the Bull River Power and Light Company on Bull River, near Fort Steele. The enterprise involves the production of over 10,000 horsepower of electrical energy from the river.

Lewiston, Ida.—Development of power on the Clearwater River at Lewiston is at last to become a reality. Work has begun on a power plant within the city limits of Lewiston, which will eventually develop more than 50,000 horsepower. The North Coast Power Company, represented in this field by Engineer Frank McKean and George W. Tannahill, is the concern undertaking the work, and it is understood that the power so developed is to be used for an extensive system of interurban lines connecting Lewiston and vicinity with Clarkston, Asotin, Anatone, Cloverland, and Pomeroy, and extending into the Craig Mountain timber belt and to the cement, lime, and coal deposits on the Snake River above Lewiston.

Tacoma, Wash.—With the granting of a franchise by the county commissioners to Donald Fletcher, the electric power plant to be established on the Teanaway River, at Cle Elum, in Kittitas County, is expected to be a reality. The franchise calls for the beginning of work within two years, a completion of one-fourth of it in five years, and operation in ten years. By the franchise, permission is granted to Mr. Fletcher to string his electric cable along all of the roads in Pierce County. The line will also have to pass through King and Kittitas Counties. Mr. Fletcher's plan is to build his power plant below the government dam on the Teanaway River, where he believes that at least 110,000 horsepower can be developed. The current brought from the river will be used in supplying light and power in Buckley, Puyallup, Sumner, and Tacoma, and other cities in Pierce and King Counties, according to present plans.

ELECTRIC RAILROADS.

Columbus, Mont.—There will be an electric railroad built between Columbus and Cooke City, a distance of sixty miles, and work on the proposition will commence within the next two months.

Helena, Mont.—It is possible that the Helena Light and Railway Company will expend \$70,000 in constructing a new car line to the fair grounds this summer, and in purchasing cars and equipment for handling the crowds which attend the annual State fair.

Turner, Ore.—The Oregon Electric has just finished its survey from Salem to this place, and Turner people confidently expect an electric railroad in a short time. There is an impression that the road will continue

on south to Albany or that a branch line will be built east into the Stayton neighborhood.

Okanogan, Wash.—The financial aid of English capitalists, who will take over an issue of \$3,000,000 in bonds at 85 cents net, practically has assured the construction of the Okanogan electric railway, a road 75 miles long, which will pass directly through the Okanogan country, according to A. M. Dewey, who has offices in the Empire State Building, Spokane.

Woodburn, Ore.—A standard-gauge electric railroad will be built by the Valley Railway Company from West Woodburn to Woodburn, three miles, and thence through Monitor to Scotts Mills, and on up into the foothills to Wilhoit Springs, a health resort in the Cascades. A branch will be extended from Monitor to Silverton. Construction work will be commenced May 15.

Portland, Ore.—An electric railway company organized by Portland and Seattle men will build a line from Condon to Bend, crossing the John Day River and securing power from that stream, also erecting a dam 200 feet high in the Deschutes River and developing power there. Among those who are engineering the deal are Dr. H. I. Keeney, George C. Mason, and Mark W. Gill.

Pendleton, Ore.—An electric line from Pendleton to Irrigon, touching at Echo, Fosters, Hermiston, and Umatilla, and tapping all that vast section of country now being reclaimed by the government and other private irrigation concerns, is one of the plans of Dr. H. W. Coe, of Portland, who is promoting lands under the Furnish project. A preliminary survey has already been made. The proposed road will be a little more than 50 miles long. It will also eventually be connected with the Milton-Walla Walla electric line.

INCORPORATION.

Santa Barbara.—The Monterey Oil and Development Company has been incorporated with a capital stock of \$100,000 by Santa Maria people.

Bakersfield.—The Golden Gate Oil Company has been incorporated with a capitalization of \$500,000. The directors are E. D. Gillette, G. R. Neill, W. H. P. McDonald, H. Lucas, and M. S. Platz.

Santa Ana.—Articles of incorporation have been filed by the Consolidated Petroleum Corporation. The incorporators are A. W. Casey, K. A. Snyder, James McDonald, and others, all of Los Angeles. The corporation is capitalized at \$1,000,000.

Modesto.—Articles of incorporation have been filed by the Tuolumne Water and Power Company. The capital stock is \$1,000,000. The incorporators are Warren Gregory and H. H. Rolfe, of San Francisco; Geo. Whipple, of Alameda; Winfield Dorn, of Oakland; and J. L. Lamson, of Berkeley.

TRANSPORTATION.

Douglas, Ariz.—Douglas and Bisbee are to be connected by an interurban traction line, which is being projected by Cochise County capitalists under the name of the Cochise County Electric Railroad Company. The men interested in the project are James S. Douglas, W. H. Brophy, and George H. Neale. The capital stock is \$500,000.

Walla Walla, Wash.—The County Commissioners have granted a franchise to the Mt. Hood Electric Railway Company to construct and operate an electric railway along the edge of the cemetery. The only condition attached is that the company start construction by the first of the year and have the road completed by January 1, 1910.

Los Angeles.—The City Council has passed ordinance 16,320, granting the Los Angeles Railway Company the right to construct and, for a period of 21 years, to operate and maintain a double-track electric street railroad, commencing at the intersection of San Pedro and Thirtieth Streets, thence southerly along San Pedro Street to its intersection with South Park Avenue.

Pasadena.—The City Council has agreed to turn down the request of the Pacific Electric Company for a franchise from Lake Avenue to east city limits on California Street and to turn down the proposition of the company to vacate rights to build on several streets in Pasadena, but voted to grant a franchise on East California Street for a short strip of railway between Mentor Avenue and Tournament Park, where the company has no franchise.

Klamath Falls, Ore.—The Klamath Falls Electric Railway Company is making arrangements for the extension of its street car line this summer. C.

N. Hawkins and W. K. Brown, directors of the company, will superintend the work. E. R. Reames, local manager, states that the Belt Line will likely be completed, as negotiations are now under way for enough steel for the entire circuit. The old horse car will be replaced by some kind of motor.

Portland, Ore.—The announcement has been made simultaneously with the increase of the capital stock of the Oregon Electric Company from \$2,000,000 to \$10,000,000 that the company would begin actual construction on the first of 283 miles of extension branches and laterals to the Portland-Salem electric line, which was placed in operation within the last few weeks. The roads mapped out are from Portland to Tillamook via Hillsboro; Portland to Eugene via Corvallis; Salem to Mill City, Salem to Dallas, Salem to Albany, and Albany to Cascadia.

Woodland.—T. C. Gregory, president of the Vallejo & Northern Railroad, states that he has taken up the condemnation proceedings against the Reed Orchard Company for the large tract of land which the company wishes to acquire opposite Sacramento. The promoters of the road say that they now have all the strategic positions, and a request has been filed with the attorney-general for permission to bring suit in the name of the State against the city of Sacramento, in order to obtain the freight franchise in that city, regarding which some trouble has been experienced.

Portland, Ore.—An electric railway company has been organized by Portland and Seattle capitalists, among them being Dr. H. I. Keeny, Geo. C. Mason, and Mark W. Gill, with Eastern capital also behind them, for the purpose of building an electric railway from Condon to Bend, crossing the John Day River. A dam 200 feet high will be erected in the Deschutes River and power developed there. It is proposed to tap the coal field near Madras and serve the Oregon King gold mine, controlled by Jack Edwards, near Ashwood. The concern will be known as the Portland Construction Company.

TRANSMISSION.

Oroville.—The new Humbug Valley power plant is but one of a number of plants that the Oro Water, Light and Power Company is preparing to construct.

Placerville.—James O'Brien and Robert Duncan have filed notice of the location and appropriation of 5,000 inches of water of the main fork of the Cosumnes River, for power, mining, and irrigation.

Lewiston, Idaho.—Commissioners Salsberg and Miller have reported favorably on the application of the North Coast Power Company for a tract of ground near the water plant for a period of three years, to be used as a site for a power plant.

Chihuahua, Mex.—The Aguila Amalgamated Mining Company, which has taken over eight properties in the Hostotipaquilla district in Jalisco, is planning to erect an electric power plant on Santiago River. J. Burpee Melly, of Boston, Mass., is president of the company.

Placerville.—R. H. Sterling, manager of the American River Electric Company, is at the company's power-house near this place to inspect the recent work of repairs which has been going on at the power-house for the past two months. Four Pelton water wheels have been installed at a cost, including setting, of \$10,000. The new wheels have a capacity of 6,000 horsepower and take the place of those placed five years ago. The large switchboard has also been received and other improvements made. Assistant Superintendent H. McGuirk and twenty-five men have been employed on the job.

San Francisco.—Agents of E. H. Harriman have secured valuable water rights in the Sierra Nevada and Siskiyou Mountains of California at four strategic points. Harriman is planning for the future, by means of these water rights, to generate electric power for the movement of Southern Pacific trains on various sections of its lines. The rights are located in the mountains of Kern County, east of Bakersfield; in Fresno County, east of Fresno; in El Dorado County, on the Rubicon River, and on the Klamath River, in Siskiyou County. It is stated that fully \$15,000,000 ultimately will be spent in developing power at these points. The power generated on the Rubicon River will operate Southern Pacific trains between Sacramento and Reno, Nevada. When this electric power plant is completed, Harriman will have finished his proposed 35,000-foot tunnel in the Sierra Nevada Mountains.

ILLUMINATION.

Porterville.—Z. G. Peck, representing the firm of Bennison & Bruntion, Los Angeles contractors, has asked for a franchise to lay gas pipes and mains.

Portland, Ore.—The County Commissioners have granted a franchise to C. L. Pritchard for an electric light system complete for the town of Washougal.

East Newport.—Test work has commenced on the natural gas locations in the big asphaltum beds just across the bay from this point, which were filed on recently by W. W. Wilson, P. T. Evans, and Archie Sharpe, of Riverside, and Lew W. Wallace, of Newport. If the supply of natural gas is found to be sufficiently large to warrant it, a plant will be established and pipe lines will be run to Newport and East Newport for lighting and cooking purposes.

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